

Chapter 1

Introduction System and Network Administration

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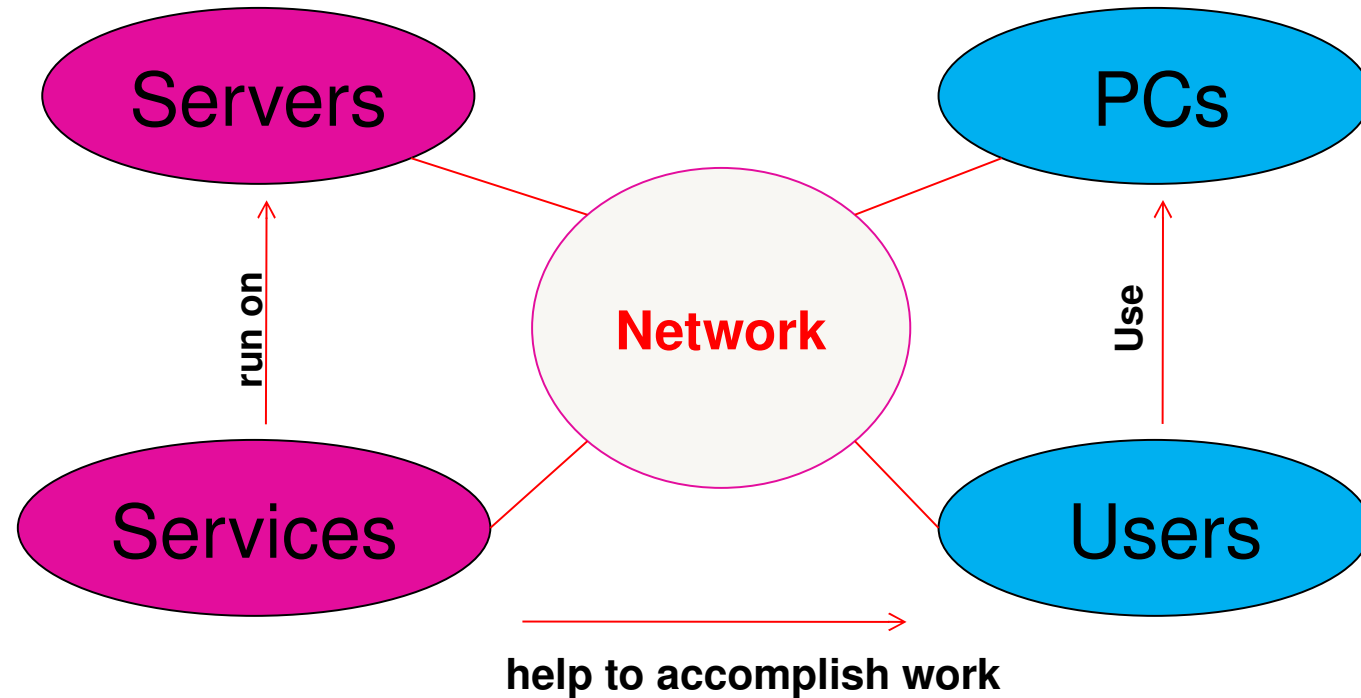


Topics

1. What is system administration?
2. What do sysadmins do?
3. Administration Challenges
4. Principles and First Steps
5. Organizations and Certifications
6. Maturity and Complexity
7. SAGE Code of Ethics
8. X-Windows Operating Systems

What is a system?

System: An organized collection of computers interacting with a group of users looking for services to accomplish work.



System State

System policy: **specification** of a system's configuration and its acceptable usage.

System state $S(t)$: the **current** configuration (files, kernel, memory or CPU usage) of a system.

Ideal states $S^*(t)$: **states of the system that match the system policy**. Over time, the system state **shifts** away from the ideal state.

System administration: modifying the system to bring it closer to $S^*(t)$.

What do sysadmins do?

Small org: sysadmin can be entire IT staff

- Phone support
- Order and install software and hardware
- Fix anything that breaks from phones to servers
- Develop software

Large org: sysadmin is one of many IT staff

- Specialists instead of “jack of all trades”
- Database admin, Network admin, Fileserver admin, Help desk worker, Programmers, Logistics

What do sysadmins do? ...

1. Add and remove users
2. Add and remove hardware
3. Perform backups
4. Install new software systems
5. Troubleshooting
6. System monitoring
7. Auditing security
8. Help users
9. Communicate

All are common activities for sysadmin.

User Account Management

Creating user accounts

- Consistency requires automation
- Startup (dot) files

Removing user accounts

- Consistency requires automation
- Many accounts across different systems

Namespace management

- Usernames and UIDs
- Multiple namespaces or SSI?



Hardware Management

Adding and removing hardware

- Configuration, cabling, etc.

Purchase

- Evaluate and purchase servers + other hardware

Capacity planning

- How many servers? How much bandwidth, storage?

Data Center management

- Power, environment (cooling, fire alarm)

Virtualization

- When can virtual servers be used vs. physical?



Data Backups

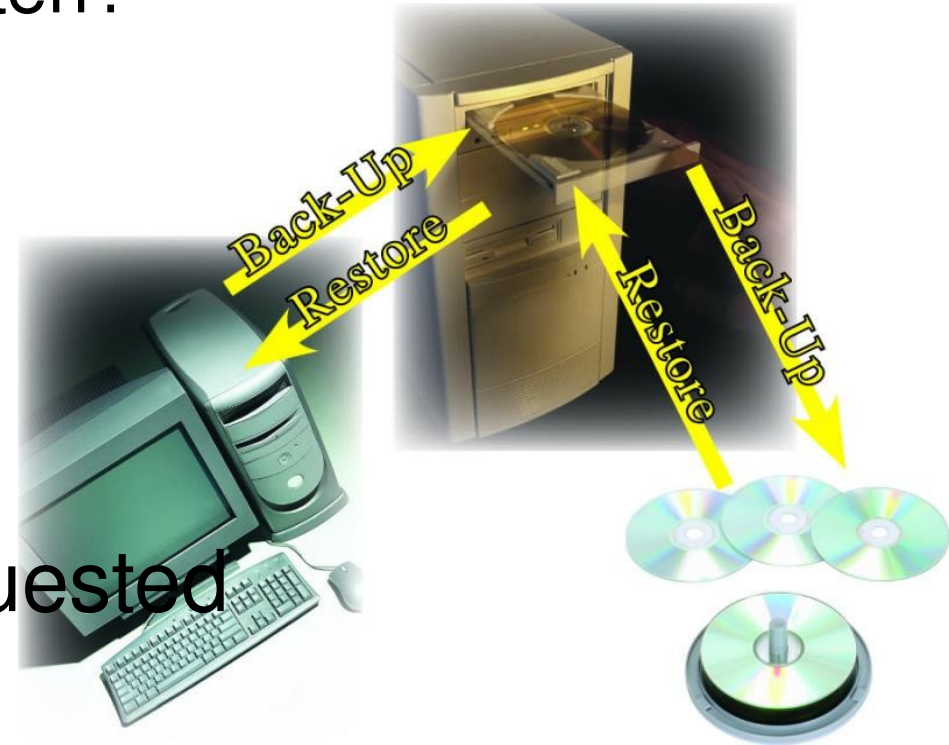
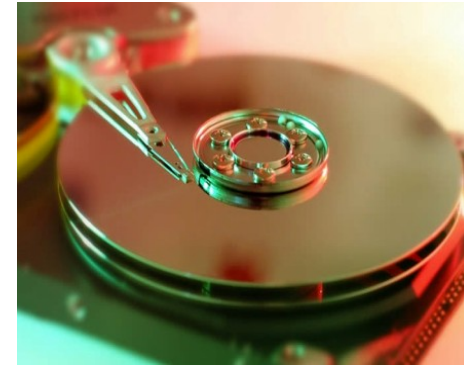
Backup strategy and policies

- Scheduling: when and how often?
- Capacity planning
- Location: on-site vs. off-site.

Monitoring backups

- Checking logs
- Verifying media

Performing restores when requested



Software Installation

Evaluation of software

Automated consistent OS installs

- Desktop vs. server OS image needs.

Installation of software

- Purchase, find, or build custom software.

Managing software installations

- Distributing software to multiple hosts.
- Managing multiple versions of a software pkg.

Patching and updating software



System Monitoring

Automatically monitor systems for

- Problems (disk full, error logs, security)
- Performance (CPU, mem, disk, network)

Security

- Passwords
- Break-ins

Capacity

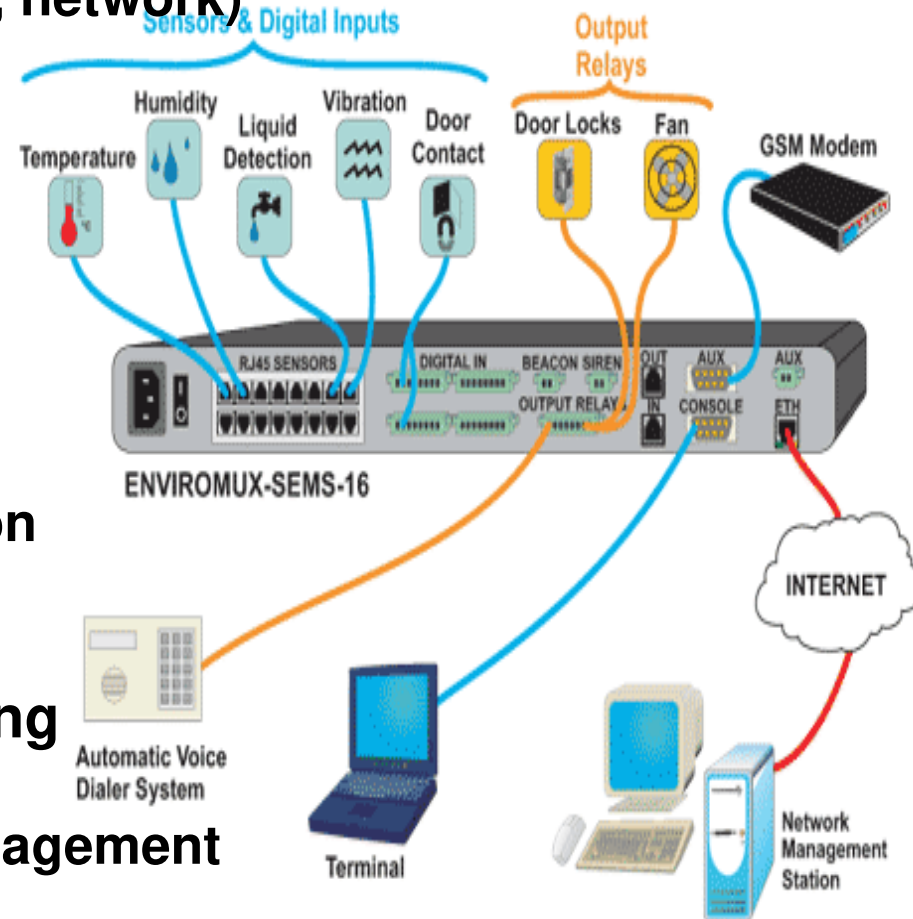
- Disk, RAM, CPU, network

System logs

- Examination
- Periodic rotation and truncation
- Archival storage (at least summaries)

Provides data for capacity planning

- Determine need for resources
- Establish case to bring to management



Troubleshooting

Problem identification, diagnosis, and resolution

- By user notification
- By log files or monitoring programs

Finding the root cause of problems

- Provide temporary solution if necessary
- Solve the root problem to permanently eliminate
 - Often quite difficult!

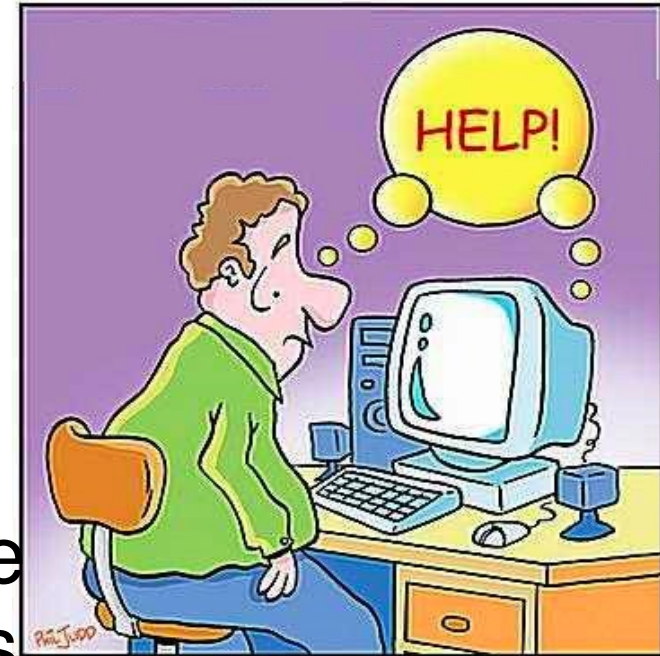
Often requires

- Broad and thorough system knowledge
- Outside experts
- Luck



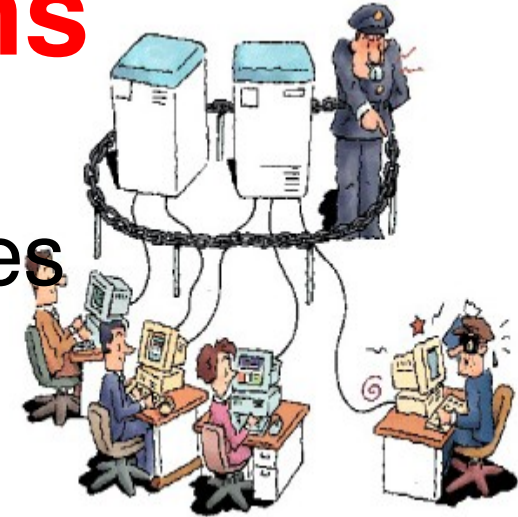
User Assistance

- Time intensive!
- Techniques
 - Help desks
 - Trouble-ticket systems
- Software availability and usage
- Software configuration settings
- Hardware usage, maintenance, and troubleshooting
- **User documentation and training**
 - Policies and procedures
- Schedule and communicate downtimes



Security Concerns

- System logging and audit facilities
 - Evaluation and implementation
 - Monitoring and analysis
 - Traps, auditing and monitoring programs
- **Unexpected or unauthorized user detection**
- Monitoring of security advisories
 - Security holes and weaknesses
 - Live exploits



Communicate

Customers

- Keep customer appraised of process
 - When you've started working on a request
 - When you make progress, need feedback
 - When you've finished
- Communicate system status
 - Uptime, scheduled downtimes, failures
- Meet regularly with customer managers

Managers

- Meet regularly with your manager
- **Write weekly status reports**

Specialized Skills

Heterogeneous Environments

Integrating **multiple-OSes**, hardware types, or network protocols, distributed sites.

Databases

SQL RDMS

Networking

Complex routing, high speed networks, voice.

Security

Firewalls, authentication, biometrics, cryptography.

Storage

Network Attached Storage (NAS), Storage Area Networks (SANs), cloud storage.

Virtualization and Cloud Computing

VMware, cloud architectures.

Practice of sysadmin

- Common practice may not be good practice
 - There are three reasons for common practice:
 - Someone did it and others followed blindly;
 - Believe it is good after careful thought;
 - An arbitrary choice had to be made;
 - Think for yourself
 - Pay attention to experts but don't automatically believe anyone
 - Every choice needs a reason

Practice of sysadmin ...

- **Good practices**
 - Look for answers in **manuals, newsgroups, and archive of mailing lists**
 - Usually **“google”** helps for most common problems
 - Use controlled trial and error for diagnosis
 - Listen to people who tell us there is a problem. It might be true
 - Write down problems and solutions in a log book, and write down experiences
 - Take responsibilities for our actions
 - Remember to tidy things up regularly
 - **After learning something new, ask yourself “How does this apply to my work?”**

System administrator (sysadmin)

- **Successful System Administrators**

- No stereotypes, “White/black Cats Theory”
- Normally if one can keep the users happy she is successful (which is very difficult)
- Life-long learning skills. One’s proud knowledge and expertise today may be obsolete tomorrow (Be prepared to be jobless if you do not keep learning)

- **Some myths for new sysadmin**

- There exists a right answer for every problem
- Things should always work in the way we expect
- Every problem should have a happy end

Challenges of System Administration

- Not only just installing system/software, also about **planning and designing an efficient community of computers**
- Design a logical and efficient network
- Easy upgrade for a large number of computers
- Decide what and where services are installed
- Plan and implement security
- Provide a comfortable environment for users
- Develop ways of fixing problems and errors
- Keep track of new technology and software

Challenges of System Administration ...

- **Need**
 - Broad knowledge of hardware and software
 - To balance conflicting requirements
 - Short-term vs. long-term needs
 - End-user vs. organizational requirements
 - Service provider vs. police model
 - To work well and efficiently **under pressure**
 - 24x7 availability
 - Flexibility, tolerance, and patience
 - Good communication skills
- **People think of sysadmins only when things don't work!**

Principles of SA

Simplicity

- Choose the simplest solution that solves the entire problem
- Work towards a predictable system

Clarity

- Choose a straightforward solution that's easy to change, maintain, debug, and explain to other SAs

Generality

- Choose reusable solutions that scale up; use open protocols

Automation

- Use software to replace human effort

Communication

- Be sure that you're solving the right problems and that people know what you're doing

Basics First

- Solve basic infrastructure problems before advanced ones

Qualities of a Successful Sysadmin

Customer oriented

- Ability to deal with interrupts, time pressure
- Communication skills
- **Service provider, not system police**

Technical knowledge

- Hardware, network, and software knowledge
- Debugging and troubleshooting skills

Time management

- Automate everything possible
- Ability to prioritize tasks: urgency and importance

First Steps to Better SA

Use a request system

- Customers know what you're doing
- You know what you're doing

Manage quick requests right

- Handle emergencies quickly
- Use request system to avoid interruptions

Policies

- How do people get help?
- What is the scope of responsibility for SA team?
- **What is our definition of emergency?**

Start every host in a known state.

Types of Sites

Small

2-10 computers, 1 OS, 2-20 users.

Small staff size requires outsourcing to obtain most specialized skills.

Midsized

11-100 computers, 1-3 OSeS, 21-100 users.

Large

100+ computers, multiples OSeS, 100+ users

Outsources to reduce costs, some specializations.

Organizations Working on SA

USENIX: Advanced Computing Systems Association

LISA: Large Installation System Administration

SAGE: System Administration Guild

LOPSA: League of Professional System Administrators

Certifications for SA

- CCNA, CCNP, CCIE (Cisco)
- cSAGE (SAGE)
- MCSA (Microsoft)
- RHCE (Red Hat)
- SCSA (Sun)
- VCP (VMware)

SAGE Job Descriptions

Novice SA

OS familiarity, help desk skills

Junior SA

Can use OS system administration tools

Intermediate SA

Understanding of distributed computing, common servers, automate small tasks, independent action

Senior SA

Understanding of scaling issues, including capacity planning, **solve problems by addressing root cause**, higher level programming abilities, write proposals for purchasing, **data center planning**, etc.

SA Maturity Model (SAMM)



1. Ad Hoc

Ad-hoc non-repeatable solutions, firefighting.

2. Repeatable

Some repeatable processes.

3. Defined

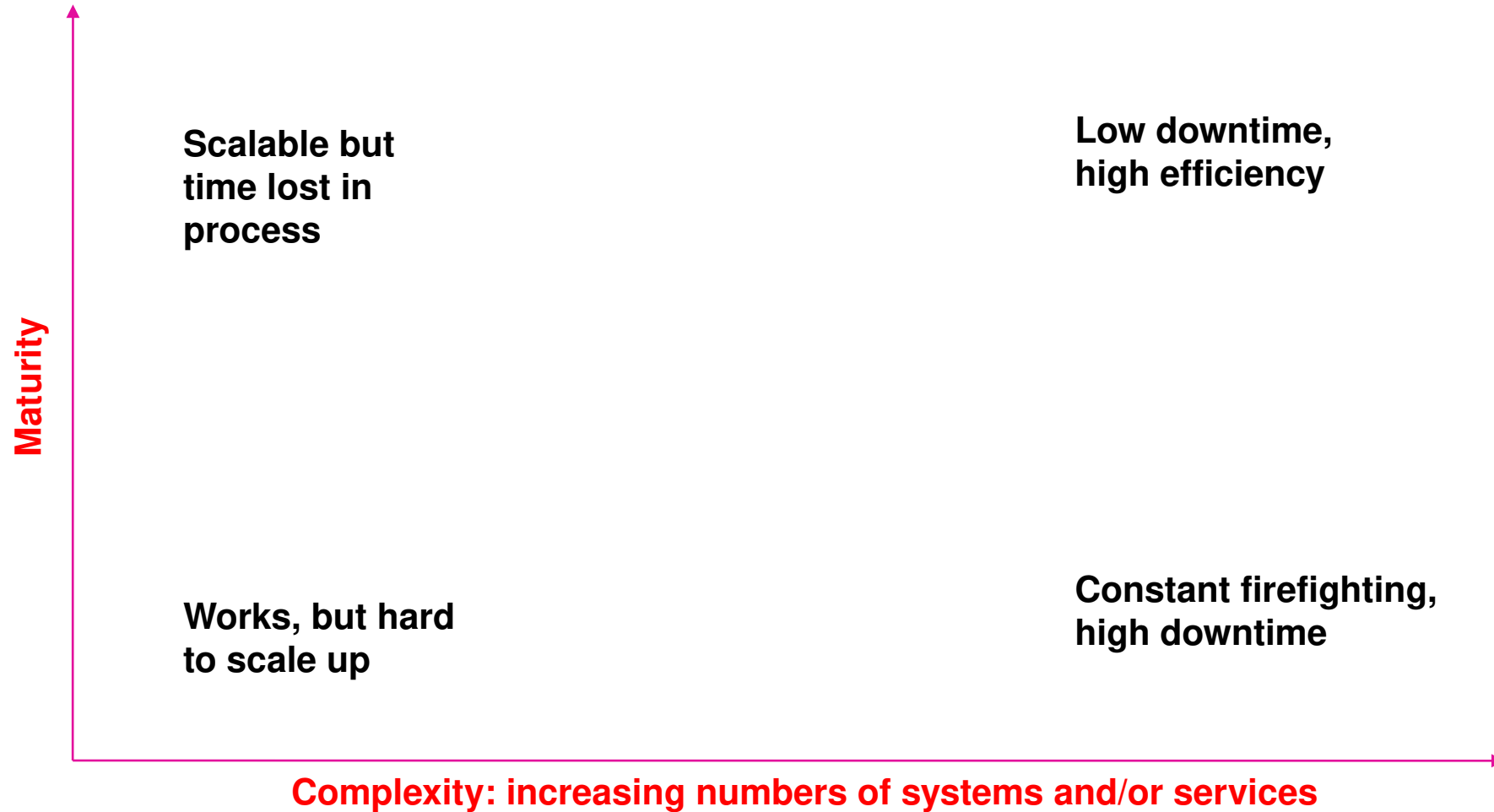
Documented standard processes

4. Managed

Process effectiveness measured, adapted.

5. Optimized

Maturity and Complexity



Tool Maturity Levels



1. Ad Hoc

OS GUI, CLI, or web administration interfaces.

2. Repeatable

Version control (RCS, SVN, GIT), request tracker

3. Defined

Automatic monitoring (Nagios, monit, god)

4. Managed

Configuration management (Automatelt, cfengine)

5. Optimized

SAGE Code of Ethics

- Professionalism
- Personal Integrity
- Privacy
- Laws and Policies
- Communication
- System Integrity
- Education
- Social Responsibility

<http://www.sage.org/ethics/>

X-Windows

- Mac Operating System
- Unix Operating System
- Linux Operating System
- etc

Which OS to learn to admin?



Brief History of UNIX

- **Originated as a research project in 1969 at AT&T Bell Labs**
 - Made available to universities (free) in 1976
- **Berkeley UNIX started in 1977 when UCB licensed code from AT&T.**
 - Berkeley Software Distribution started in 1977 with 1BSD, and ended in 1993 with 4.4BSD
 - Licensing costs from AT&T increased, so Berkeley attempted to remove AT&T code, but ran out of funds before completion.
 - Final release of AT&T-free code called 4.4BSD-Lite.
 - Most current BSD distributions (FreeBSD, NetBSD, OpenBSD) are derived from 4.4BSD-Lite.
- **Most commercial versions of UNIX (Solaris, HP-UX) are derived from the AT&T code**

What is Linux?

much is courtesy of www.kernel.org

- Linux is a clone of the operating system Unix, written by a loosely-knit team of hackers across the Net.
- Like any modern full-fledged Unix, Linux includes:
 - true multitasking
 - proper memory management and virtual memory
 - shared libraries and shared copy-on-write executables
 - TCP/IP networking
- Linux really refers to the kernel
 - Most commands and apps are really separate programs, not specific to Linux, and often are part of the Free Software Foundation's GNU project.
- Linux was first developed for 32-bit x86-based PCs (386 or higher). These days it also runs on dozens of other processors.

What is Linux? Cont...

- Linux stands for Linus' Unix
- It is the **kernel** of an operating system.
- **Linux** is an open-source, UNIX-like operating system created by Linus Torvalds in 1991.
- Known for being secure, customizable, and widely used in servers, desktops, and embedded systems.
- **Key Features:** Multitasking, multi-user support, portability, and stability.
- **Kernel:** The core of the Linux OS, managing hardware and system processes.

Brief History of Linux

- Created as a **personal project** (and still controlled) by Linus Torvalds, a Finnish graduate student, in 1991
- Conceived as an offshoot of Minix (a model OS)
 - Not derived from AT&T or BSD UNIX code
- Red Hat (one of many Linux vendors) founded in 1993
- Kernel v1.0 released 1994
- **Most recent (Jan 2016) kernel release is 4.4?**

Linux Distributions and UIs

- **Linux Distribution (Distro):** A complete OS package that includes the Linux kernel, GNU utilities, a package manager, and applications.
- **Popular Distros:**
 - **Ubuntu:** User-friendly, great for beginners.
 - **Fedora:** Cutting-edge features, preferred by developers.
 - **Debian:** Stable, extensive package support.
 - **CentOS/RHEL:** Enterprise-grade, popular for servers.
 - **Arch Linux:** Highly customizable, preferred by advanced users.

Basic Benefits of Using Linux

- **Open Source:** Free and open to modification.
- **Security:** Fewer vulnerabilities than other OS.
- **Flexibility:** Highly customizable for any type of use.
- **Performance:** Efficient resource usage, especially for servers

Linux Operations Review

1. File Management Commands (for navigating and manipulating files)

- **ls**: Lists the files and directories in the current directory.
Example: `ls -l` shows a detailed (long) listing with permissions, ownership, and size.
- **cd**: Changes the current directory to a specified path.
Example: `cd /home/user/Documents` changes to the "Documents" directory.
- **cp**: Copies files or directories from one location to another.
Example: `cp file.txt /home/user/backup/` copies "file.txt" to the "backup" directory.
- **mv**: Moves files or directories, or renames them.
Example: `mv file.txt /home/user/backup/` moves "file.txt" to the "backup" directory. If in the same directory, it acts as a rename.
- **rm**: Deletes files or directories.
Example: `rm file.txt` deletes "file.txt". Adding `-r` (recursive) is needed to delete directories and their contents (e.g., `rm -r myfolder`).

Cont..

2. Process Management Commands (for managing running processes)

- **ps**: Displays a snapshot of current processes.
 - Example: `ps aux` shows all running processes, their user, CPU, memory usage, etc.
- **top**: Continuously displays running processes in real-time, allowing you to monitor CPU, memory, and other resource usage.
 - You can press `q` to quit the `top` interface.
- **kill**: Terminates a specific process by its Process ID (PID).
 - Example: `kill 1234` sends a termination signal to the process with PID 1234.
`kill -9 1234` sends a forceful kill signal.

Cont..

3. Permissions Management Commands (for controlling access to files and directories)

- **chmod**: Changes file or directory permissions for users, groups, and others.
 - Example: `chmod 755 script.sh` gives read, write, and execute permissions to the owner, and read and execute permissions to group and others.
- **chown**: Changes the ownership of a file or directory.
 - Example: `chown user:group file.txt` sets "user" as the owner and "group" as the group owner of "file.txt".
- **User Groups**: Groups in Linux allow permissions to be set for multiple users at once. Commands like `usermod -aG groupname username` can add a user to a group.

Cont..

4. Package Management Commands (for installing, updating, and managing software)

- **Debian-based (e.g., Ubuntu)**

- **apt-get**: Used for package management tasks such as installing, upgrading, and removing software.
 - Example: `sudo apt-get install package-name` installs a package.
- **apt**: A newer, user-friendly command for managing packages, often with simplified syntax.
 - Example: `sudo apt update` updates the package list, and `sudo apt upgrade` upgrades installed packages.

- **RedHat-based (e.g., Fedora)**

- **yum**: Used for package management on older RedHat-based systems.
 - Example: `sudo yum install package-name` installs a package.
- **dnf**: The newer package manager replacing yum, with faster performance and dependency handling.
 - Example: `sudo dnf install package-name` installs a package.

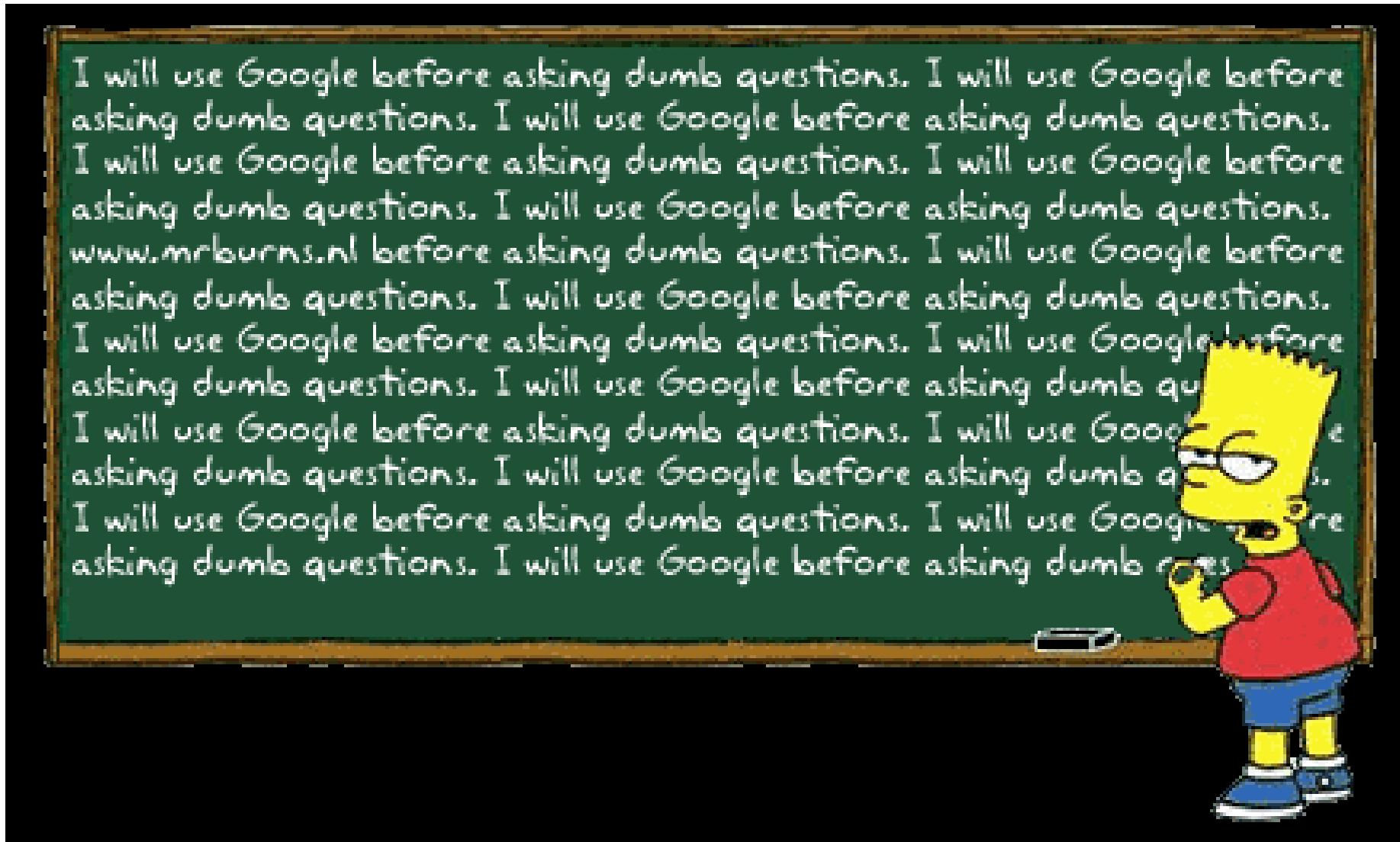
- **Arch-based (e.g., Arch Linux)**

- **pacman**: The default package manager for Arch-based systems, used for installing, upgrading, and removing packages.
 - Example: `sudo pacman -S package-name` installs a package, and `sudo pacman -R package-name` removes it.

Where to get answers?

- Linux/UNIX documentation can be found in many places
 - Manual pages (man pages, using **man** command)
 - Texinfo documents (read with **info** command)
 - HOWTOs – focused descriptions of a topic
 - Distribution-specific documentation
 - Your favorite Web search engine
 - Will typically find online versions of the above

Where to get answers?



End of Chapter 1

Thank you !